1 NAND2 Gate Design

1.1 Logic Design and Sizing

To achieve equivalent resistance to the unit inverter (NMOS 400 nm, PMOS 600 nm), the transistor sizing for the 2-input NAND gate is calculated as follows:

- **NMOS Network:** Two NMOS transistors are in series. To maintain the same pull-down resistance as a single unit NMOS, the width of each NMOS is doubled.

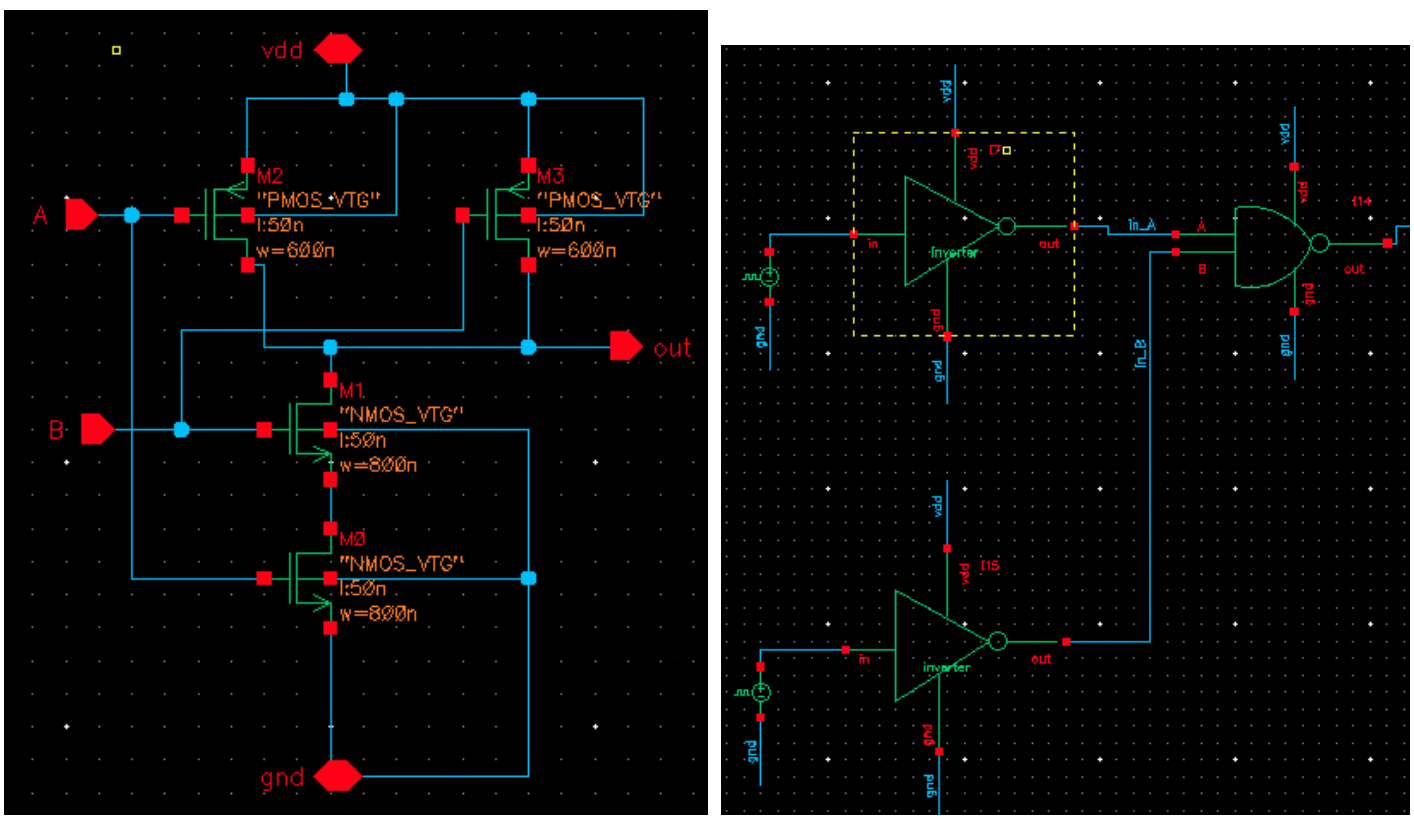
$$W_{n,NAND} = 2 \times W_{n,unit} = 2 \times 400\,\text{nm} = 800\,\text{nm}$$

- **PMOS Network:** Two PMOS transistors are in parallel. The worst-case pull-up resistance occurs when only one PMOS is on. Therefore, the width remains the same as the unit PMOS.

$$W_{p,NAND} = W_{p,unit} = 600\,\text{nm}$$

1.2 Schematic and Testbench

Figure 1 shows the schematic of the NAND2 gate and the testbench used for simulation. The testbench includes an input driving inverter and a load of 4 unit inverters (FO4) with $V_{DD} = 1.1\,\text{V}$.



(a) NAND2 Schematic. (b) NAND2 Testbench.

Figure 1: NAND2 gate design and verification setup.

1.3 Layout

The layout of the NAND2 cell is designed to match the standard cell height established by the NOR2 cell. Figure 2 displays the layout with ruler measurements indicating the dimensions. The key layout parameters are summarized in Table 1.

Table 1: NAND2 Layout Dimensions

Parameter	Value
Height	1.7175 μm
Width	0.94 μm
Area	1.6145 μm^2

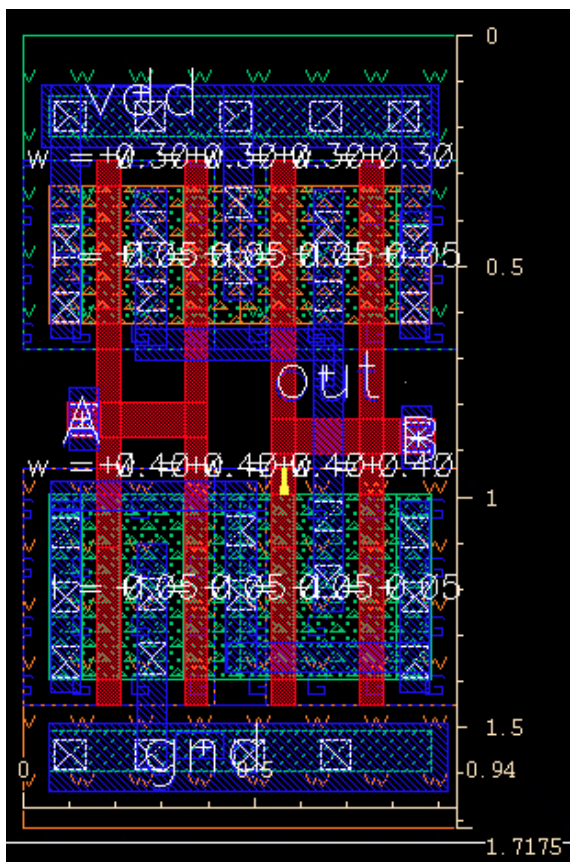
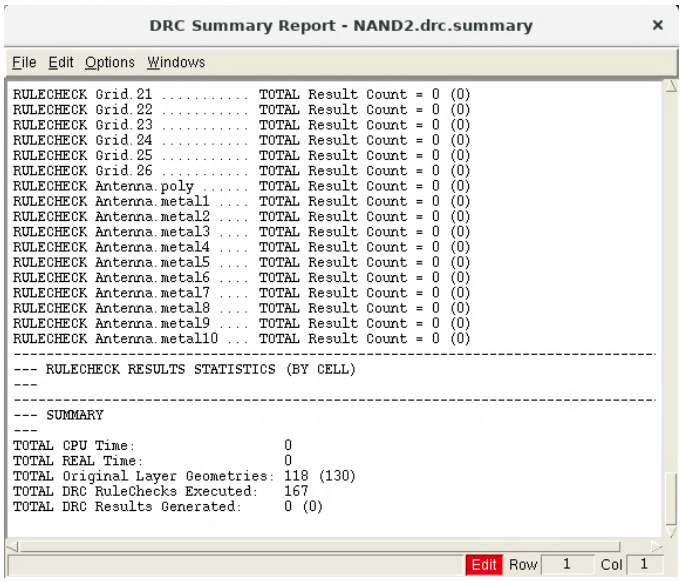


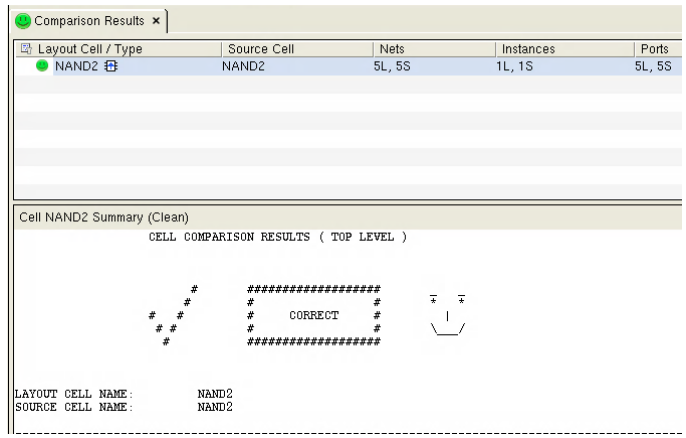
Figure 2: NAND2 Layout with ruler measurements.

1.4 DRC and LVS Verification

The design passed both Design Rule Check (DRC) and Layout Versus Schematic (LVS) checks as Figure 3 shown in below.



(a) DRC Result (Clean).



(b) LVS Result (Match).

Figure 3: Verification results for NAND2.

1.5 Extraction Summary

Figure 4 shows the post-layout RC extraction results for the NAND2 gate. The report lists the extracted parasitic resistance and capacitance for each net. Additionally, the internal node between the series PMOS transistors (identified as `net19` or similar in the source net column) is also extracted, contributing to the internal delay of the pull-up network.

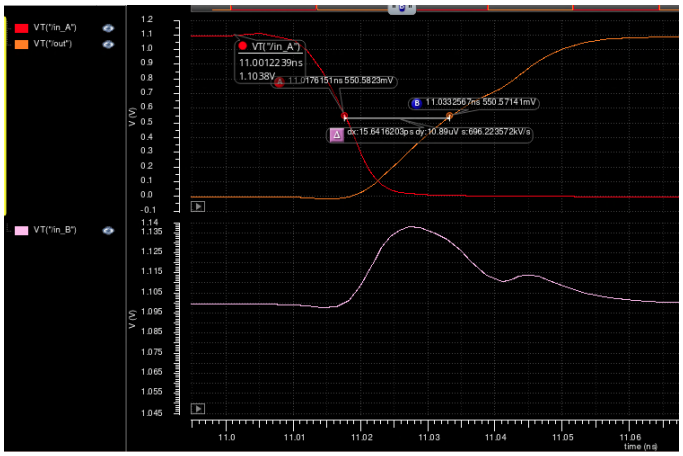
NAND2

No.	Layout Net	Source Net	R Count	C Total (F)	CC Total (F)	C+CC Total (F)
1	A	A	14	1.19365E-16	9.76858E-17	2.17051E-16
2	gnd	gnd	5	6.66299E-17	1.27290E-16	1.93920E-16
3	3	net19	10	5.84779E-17	2.39341E-16	2.97819E-16
4	out	out	10	4.60002E-17	2.06243E-16	2.52243E-16
5	B	B	15	1.22427E-16	1.09054E-16	2.31481E-16
6	vdd	vdd	12	9.50385E-17	1.32408E-16	2.27446E-16

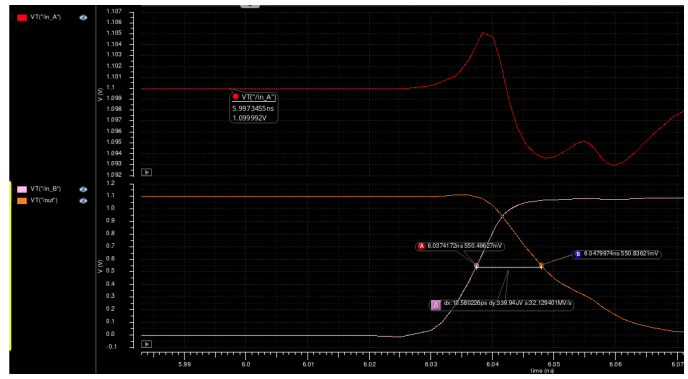
Figure 4: NAND2 Layout extraction summary.

1.6 Simulation Results

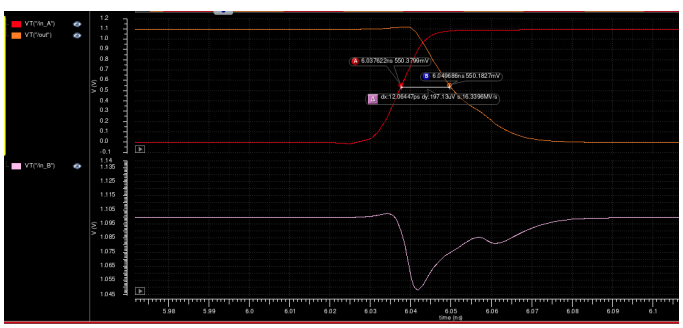
The NAND2 gate was simulated under different input transition conditions to determine the worst-case propagation delay. Figure 5 shows the transient response for four key transitions. The schematic simulation results are summarized in Table 2.



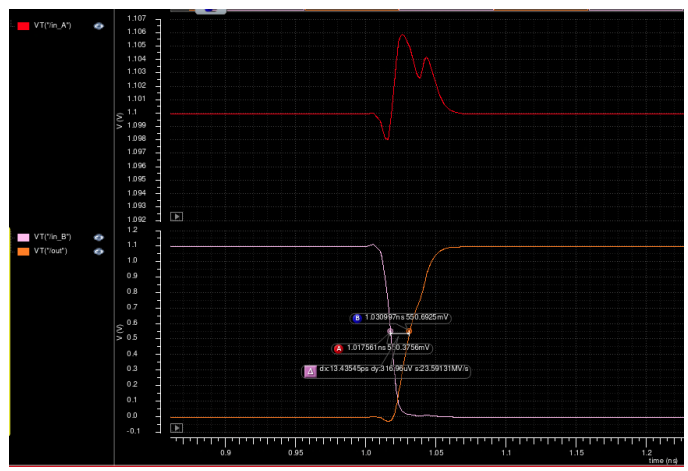
(a) Input A: 0 → 1, B=1 (Output Fall)



(b) Input B: 0 → 1, A=1 (Output Fall)



(c) Input A: 1 → 0, B=1 (Output Rise)



(d) Input B: 1 → 0, A=1 (Output Rise)

Figure 5: NAND2 schematic simulation waveforms for different input transitions.

Table 2: NAND2 Schematic Delay Analysis

Transition	Condition	Delay (ps)	Note
Output Fall (1 → 0)	A: 0 → 1 (B=1)	15.64	Worst-Case Fall
Output Fall (1 → 0)	B: 0 → 1 (A=1)	10.58	
Output Rise (0 → 1)	A: 1 → 0 (B=1)	12.06	
Output Rise (0 → 1)	B: 1 → 0 (A=1)	13.44	Worst-Case Rise

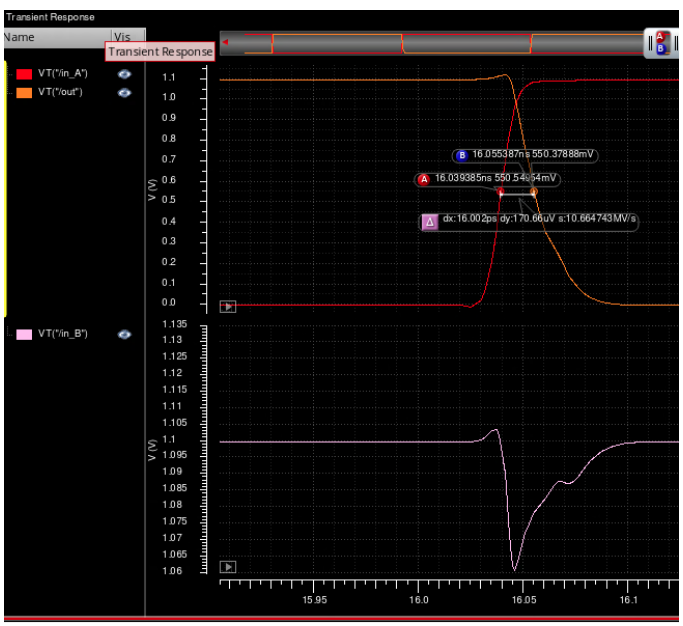
- **Fall Delay (t_{pHL}):** The worst-case delay (15.64 ps) occurs when Input A (bottom NMOS) switches (0 → 1) while B is high. This transition requires discharging both the output capacitance and the intermediate node capacitance.
- **Rise Delay (t_{pLH}):** The worst-case rise delay (13.44 ps) occurs when Input B switches (1 → 0). The parallel PMOS structure theoretically offers faster charging, but minor asymmetries make Input B slightly slower in the schematic view.

Based on the schematic logic, the series NMOS pull-down network (Fall transition) was expected to be the bottleneck due to higher resistance and internal capacitance. However, as the post-layout results will show, parasitic routing capacitance shifts this behavior significantly.

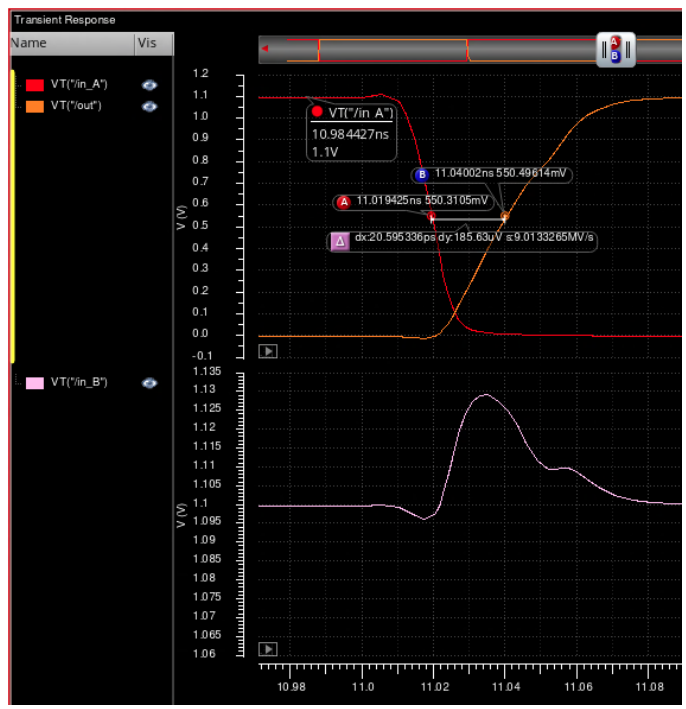
The post-layout simulation incorporates the parasitic R/C effects extracted from the layout. To rigorously determine the worst-case delay, all input transitions were simulated.

Surprisingly, the results contradict the initial intuition that the series NMOS structure would dominate the delay. Instead, the **Output Rise transition triggered by Input A** proved to be the absolute worst-case path. Figure 6 displays the transient response for the critical transitions. Table 3 summarizes the delay performance.

- **Fall Delay (t_{pHL}):** 16.00 ps (Input A). This is very close to the schematic value (15.64 ps), indicating that the layout minimized parasitic capacitance on the critical intermediate node of the NMOS stack.
- **Rise Delay (t_{pLH}):** 20.60 ps (Input A). This represents a **significant increase (+70.8%) compared to the schematic (12.06 ps)**. This anomaly suggests that the layout routing for Input A's PMOS path introduces substantial parasitic capacitance, making the "parallel" charging path significantly slower than the "series" discharging path.



(a) Post-Layout Output Fall (Input A, 16.00 ps)



(b) Post-Layout Output Rise (Input A, 20.60 ps)

Figure 6: NAND2 Post-Layout Simulation Results (Input A Worst-Case).

Table 3: NAND2 Worst-Case Delay Comparison

Metric	Schematic	Extracted	% Diff
Fall Delay (t_{pHL})	15.64 ps	16.00 ps	2.3%
Rise Delay (t_{pLH})	12.06 ps	20.60 ps	70.8%

2 NOR2 Gate Design

2.1 Logic Design and Sizing

For the NOR2 gate:

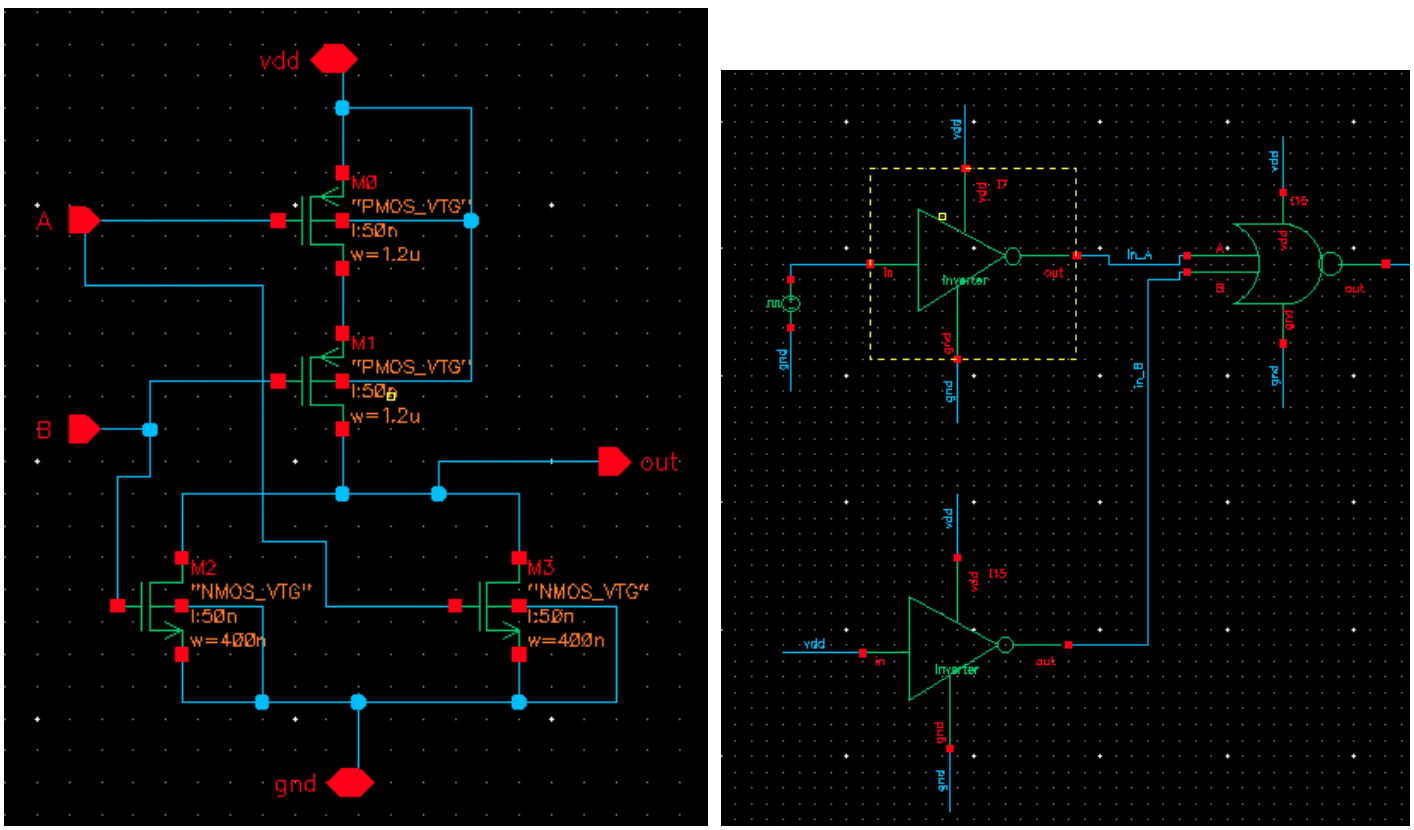
- **PMOS Network:** Two PMOS transistors are in series. To match unit resistance:

$$W_{p,NOR} = 2 \times W_{p,unit} = 2 \times 600\,\text{nm} = 1.2\,\mu\text{m}$$

- **NMOS Network:** Two NMOS transistors are in parallel. Worst case is one NMOS on:

$$W_{n,NOR} = W_{n,unit} = 400\,\text{nm}$$

2.2 Schematic and Testbench



(a) NOR2 Schematic. (b) NOR2 Testbench.

Figure 7: NOR2 gate design and verification setup.

2.3 Layout

The layout of the NOR2 gate is shown in Figure 8, with key dimensions and area summarized in Table 4.

The NOR2 cell design serves as the height bottleneck for the entire standard cell library (NAND2, NOR2, MUX2). This is primarily due to the PMOS network structure, where two transistors are connected in series. To match the pull-up resistance of the unit inverter, the width

of each PMOS transistor must be doubled ($W_{p,NOR} = 2 \times W_{p,unit} = 1.2\,\mu$m). Although multi-finger layout techniques were employed to optimize the shape, the substantial active area required for the 1.2 μm PMOS devices dictates the minimum vertical spacing between the VDD and GND rails, thereby determining the fixed cell height for all other gates in the library.

Table 4: NOR2 Layout Dimensions

Parameter	**Value**
Height	1.7175 μm
Width	0.94 μm
Area	1.61445 μm^2

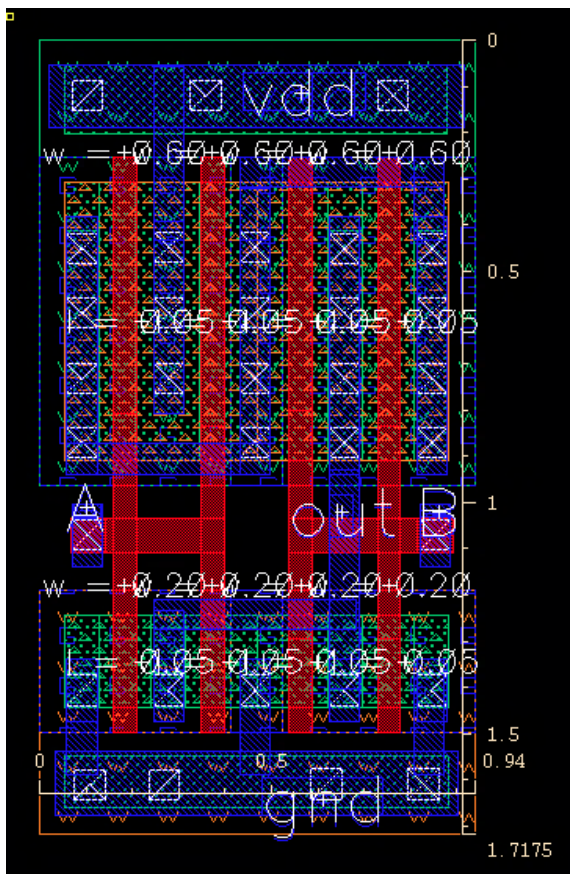
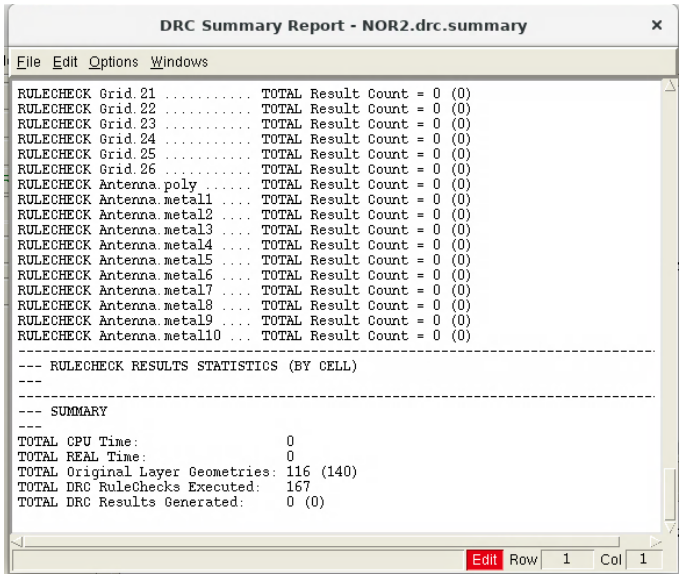
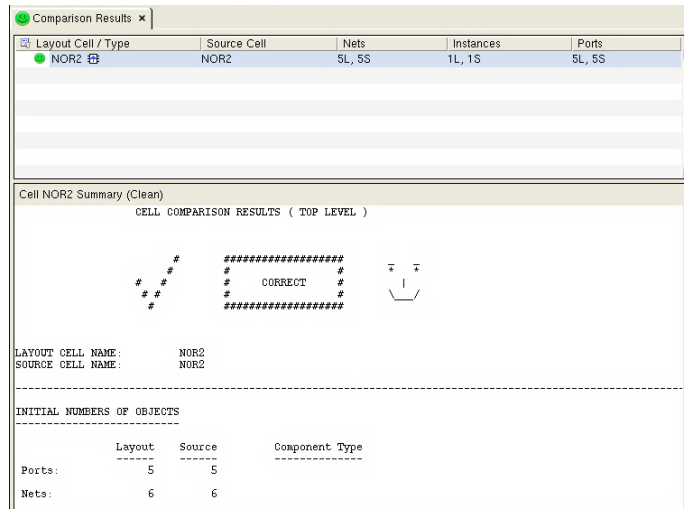


Figure 8: NOR2 Layout with ruler measurements.

2.4 DRC and LVS Verification



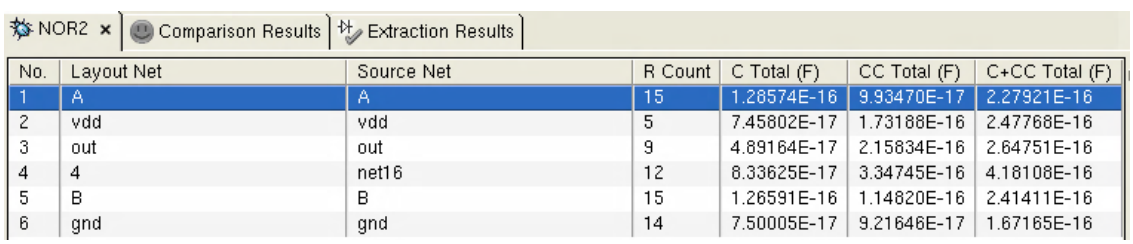
(a) DRC Result (Clean).



(b) LVS Result (Match).

Figure 9: Verification results for NOR2.

2.5 Extraction Summary



NOR2 | Comparison Results | Extraction Results

No.	Layout Net	Source Net	R Count	C Total (F)	CC Total (F)	C+CC Total (F)
1	A	A	15	1.28574E-16	9.93470E-17	2.27921E-16
2	vdd	vdd	5	7.45802E-17	1.73188E-16	2.47768E-16
3	out	out	9	4.89164E-17	2.15834E-16	2.64751E-16
4	4	net16	12	8.33625E-17	3.34745E-16	4.18108E-16
5	B	B	15	1.26591E-16	1.14820E-16	2.41411E-16
6	gnd	gnd	14	7.50005E-17	9.21646E-17	1.67165E-16

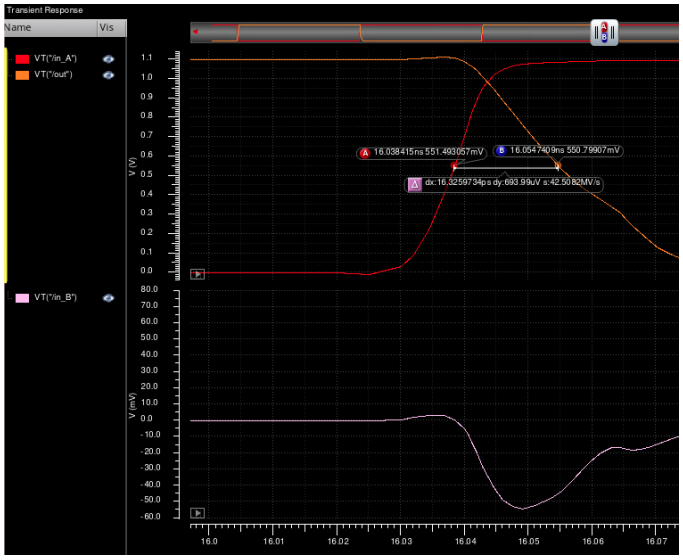
Figure 10: NOR2 Layout extraction summary.

2.6 Simulation Results

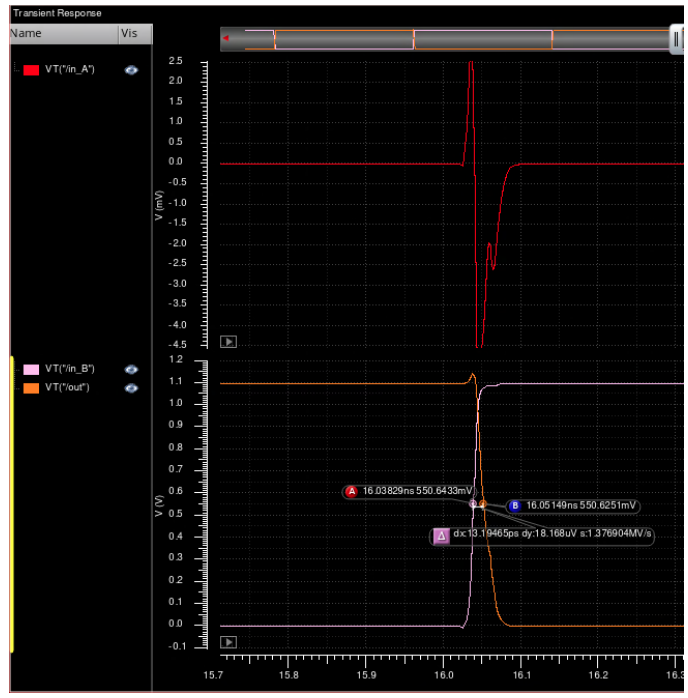
The NOR2 gate was simulated under different input transition conditions to determine the worst-case propagation delay. Figure 11 shows the transient response for four key transitions. The schematic simulation results are summarized in Table 5.

Table 5: NOR2 Schematic Delay Analysis

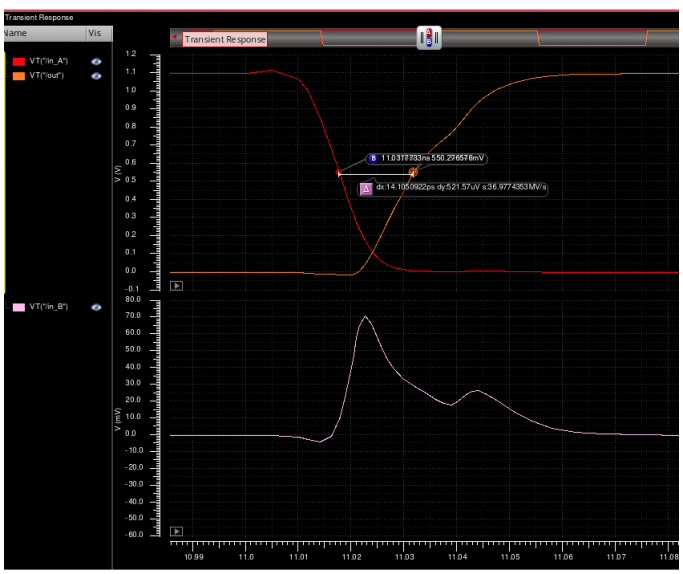
Transition	Condition	Delay (ps)	Note
Output Fall ($1 \rightarrow 0$)	A: $0 \rightarrow 1$ (B=0)	16.325	Worst-Case Fall
Output Fall ($1 \rightarrow 0$)	B: $0 \rightarrow 1$ (A=0)	13.194	
Output Rise ($0 \rightarrow 1$)	A: $1 \rightarrow 0$ (B=0)	14.105	Worst-Case Rise
Output Rise ($0 \rightarrow 1$)	B: $1 \rightarrow 0$ (A=0)	11.438	



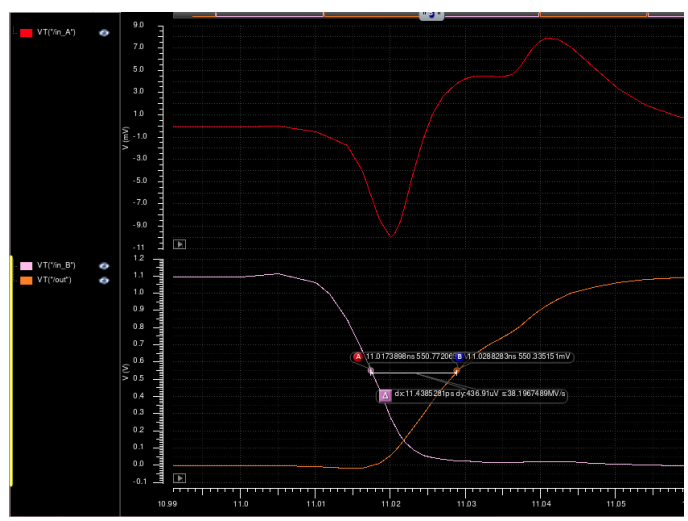
(a) Input A: $0 \rightarrow 1$, B=0 (Output Fall)



(b) Input B: $0 \rightarrow 1$, A=0 (Output Fall)



(c) Input A: $1 \rightarrow 0$, B=0 (Output Rise)



(d) Input B: $1 \rightarrow 0$, A=0 (Output Rise)

Figure 11: NOR2 schematic simulation waveforms for different input transitions.

- **Rise Delay (t_{pLH}):** The worst-case rise delay (14.105 ps) occurs when Input A (top PMOS) switches ($1 \rightarrow 0$) while B is low. This transition involves charging both the output capacitance and the intermediate node capacitance through the series PMOS stack, which theoretically presents the highest resistance path.

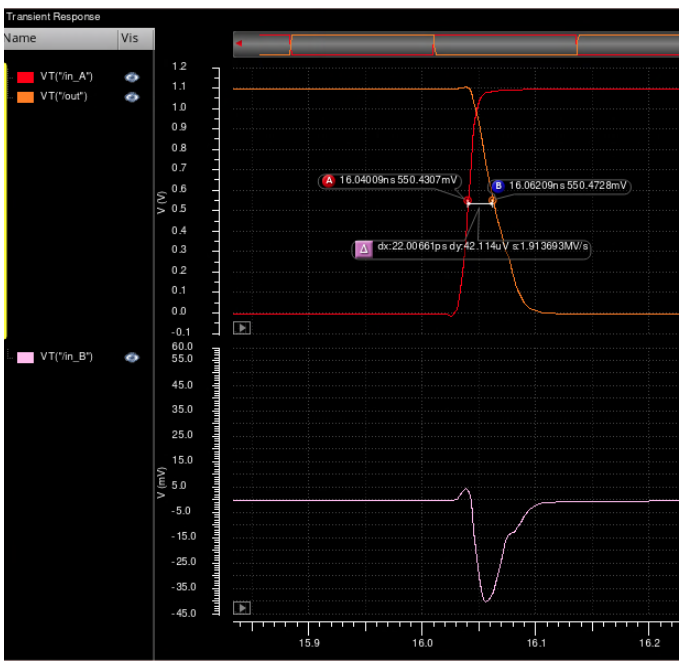
- **Fall Delay (t_{pHL}):** The parallel NMOS structure theoretically provides symmetric and faster pull-down paths. However, simulation results show that Input A switching ($0 \rightarrow 1$) yields a delay (16.325 ps) that is unexpectedly larger than the series pull-up delay.

Based on the schematic structure, the **series PMOS pull-up network (Rise transition)** was expected to be the critical bottleneck due to the stacked resistance. However, the **pre-layout simulation indicates that the NMOS pull-down path (specifically Input A) is currently slower**.

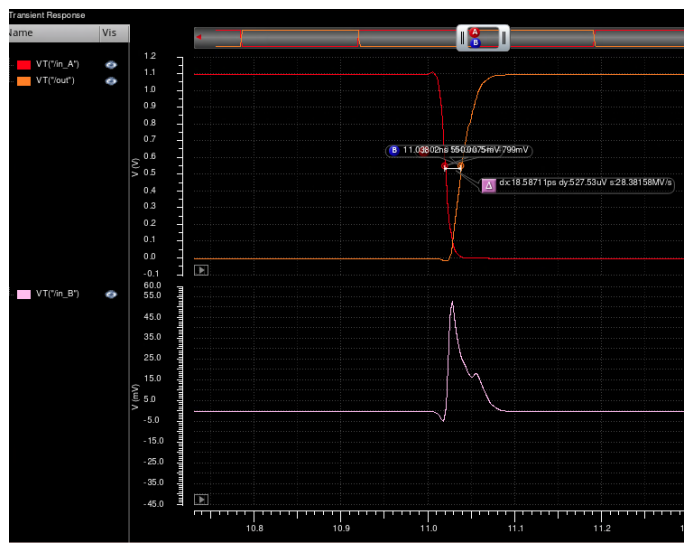
Contrary to the expectation that the series PMOS stack would dominate, the post-layout results confirm that the Fall transition remains the worst-case delay. This suggests that the diffusion capacitance of the parallel NMOS transistors, combined with the layout routing for Input A, introduces significant parasitic loading that outweighs the resistance penalty of the series PMOS stack.

Figure 12 displays the transient response of the extracted NOR2 gate for both edges. Table 6 summarizes the delay performance comparison.

- **Fall Delay (t_{pHL}):** Increases to 22.01 ps. This remains the absolute worst-case delay, confirming that the parallel NMOS diffusion capacitance is the dominant factor in this layout.
- **Rise Delay (t_{pLH}):** Increases to 18.59 ps. While slower than the schematic, it is still faster than the fall delay, disproving the initial intuition that the series PMOS would be the bottleneck.



(a) Post-Layout Output Fall (22.01 ps)



(b) Post-Layout Output Rise (18.59 ps)

Figure 12: NOR2 Post-Layout Simulation Results (Input A worst-case).

Table 6: NOR2 Worst-Case Delay Comparison (Input A)

Metric	Schematic	Extracted	% Diff
Fall Delay (t_{pHL})	16.325 ps	22.006 ps	34.8%
Rise Delay (t_{pLH})	14.105 ps	18.587 ps	31.8%

3 MUX2 Design

3.1 Logic Design and Sizing

The 2:1 Multiplexer (MUX2) was designed using **Transmission Gate (TG) Logic** to minimize transistor count and area while maintaining high performance. The schematic consists of one inverter (to invert the Select signal) and two transmission gates.

- **Sizing Strategy:** All transistors were sized using the unit inverter dimensions to maintain consistency and fit within the standard cell height defined by the NOR2 cell.
- **Inverter for Select Signal:**

$$W_n = 400\,\text{nm}, \quad W_p = 600\,\text{nm}$$

- **Transmission Gates:** Both transmission gates use the same sizing to ensure balanced rise and fall characteristics and equivalent drive strength:

$$W_{n,TG} = 400\,\text{nm}, \quad W_{p,TG} = 600\,\text{nm}$$

3.2 Schematic

Figure 13 shows the schematic of the MUX2 gate.

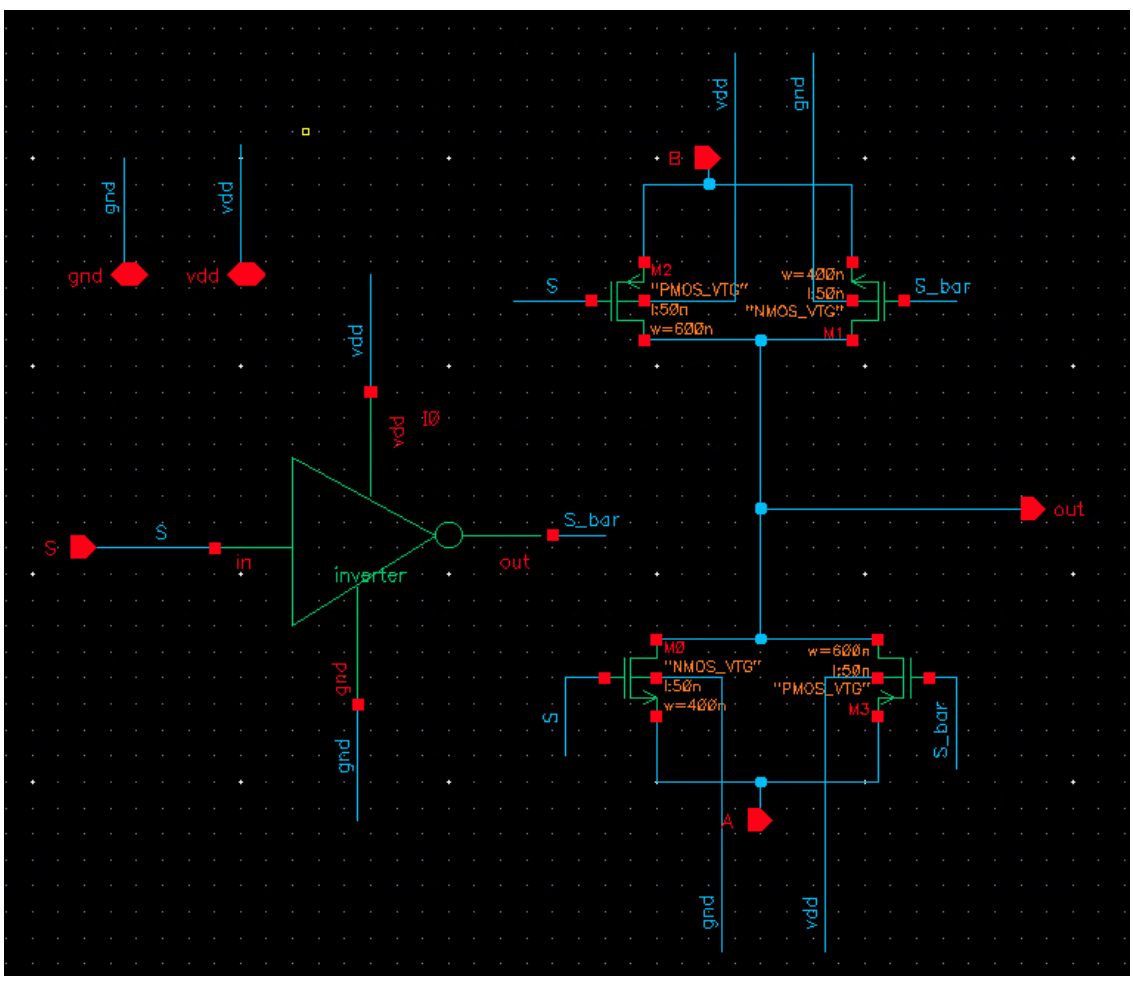


Figure 13: MUX2 Schematic (Transmission Gate Implementation).

3.3 Layout

To ensure the MUX2 cell integrates seamlessly into the standard cell library, the layout height was strictly constrained to match the NOR2 and NAND2 cells. The N-well and P-well layers were extended (or added) to ensure consistent well boundaries and connectivity across the entire library. Key dimensions are summarized in Table 7.

Table 7: MUX2 Layout Dimensions

Parameter	Value
Height	1.7175 μm
Width	1.3175 μm
Area	2.2628 μm^2

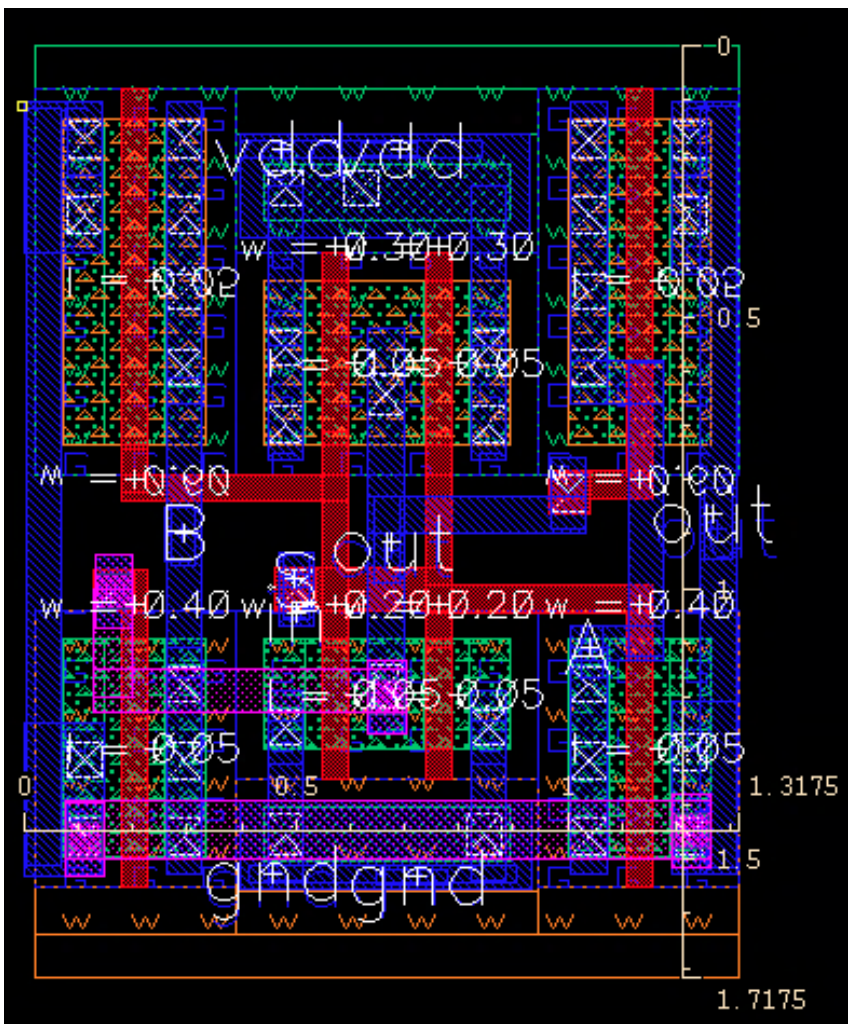
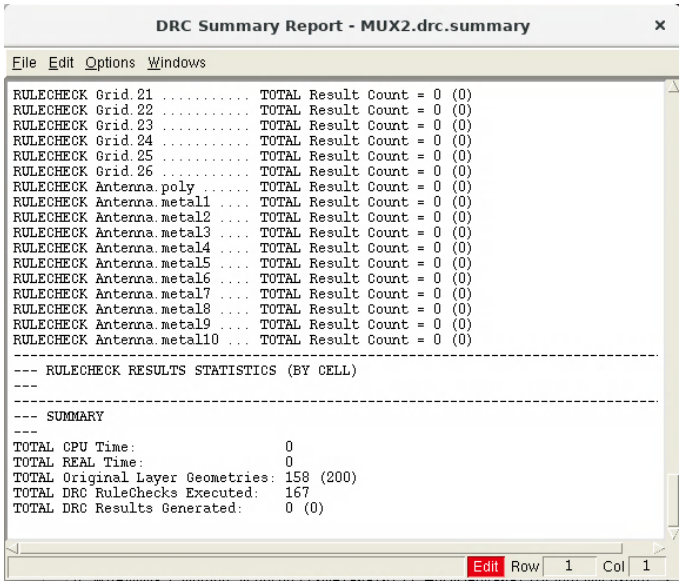
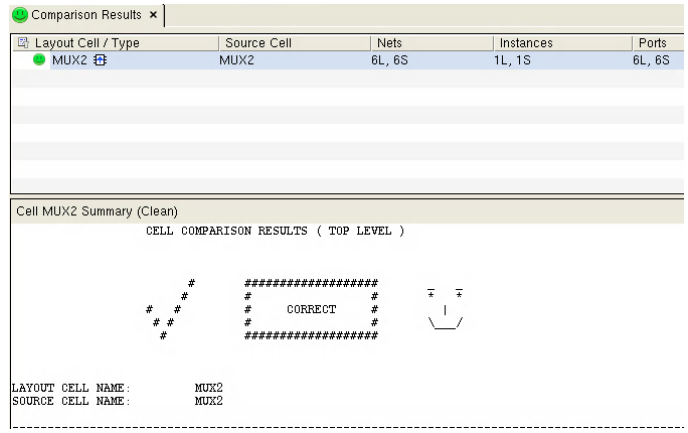


Figure 14: MUX2 Layout with ruler measurements.

3.4 DRC and LVS Verification



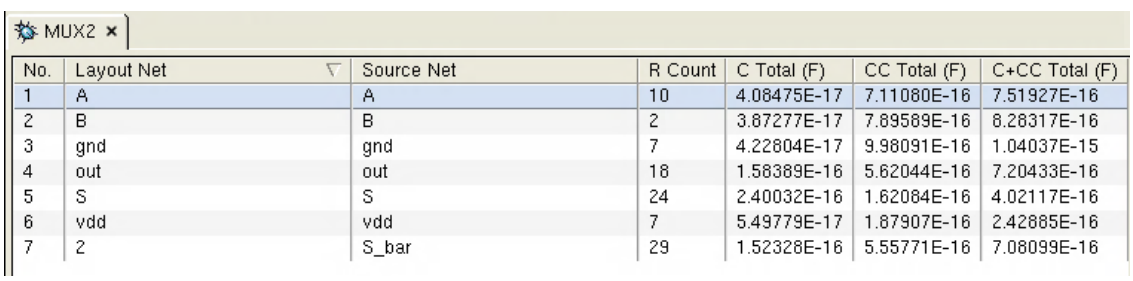
(a) DRC Result (Clean).



(b) LVS Result (Match).

Figure 15: Verification results for MUX2.

3.5 Extraction Summary



No.	Layout Net	Source Net	R Count	C Total (F)	CC Total (F)	C+CC Total (F)
1	A	A	10	4.06475E-17	7.11080E-16	7.51927E-16
2	B	B	2	3.87277E-17	7.89589E-16	8.28317E-16
3	gnd	gnd	7	4.22804E-17	9.98091E-16	1.04037E-15
4	out	out	18	1.58389E-16	5.62044E-16	7.20433E-16
5	S	S	24	2.40032E-16	1.62084E-16	4.02117E-16
6	vdd	vdd	7	5.49779E-17	1.87907E-16	2.42885E-16
7	2	S_bar	29	1.52328E-16	5.55771E-16	7.08099E-16

Figure 16: MUX2 Layout extraction summary.

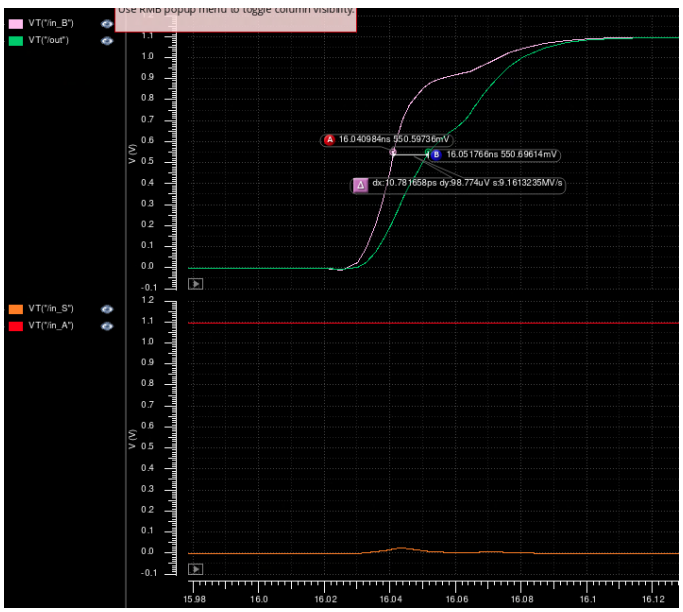
3.6 Simulation Results

The MUX2 gate was simulated to characterize the propagation delay from both the Data inputs (A, B) and the Select input (S) to the Output. The simulation results for key transitions are summarized in Table 8. Figure 17 displays the transient response for a typical Data path transition and the worst-case Select path transition.

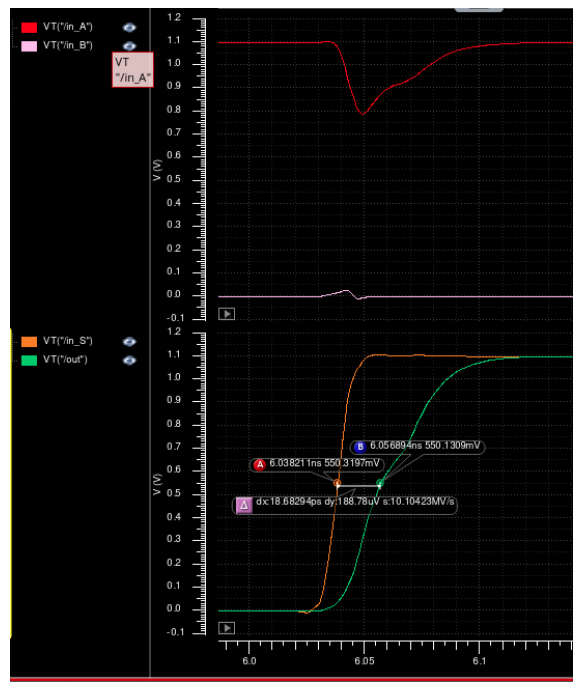
- **Data Path Delay (A/B → Out):** When the Select signal is stable, the delay is determined primarily by the on-resistance of the transmission gate and the output load. The delays are relatively fast and symmetric, ranging from approximately 9.9 ps to 10.8 ps.
- **Select Path Delay (S → Out):** The delay from the Select input is significantly larger, particularly for the 0 → 1 transition (18.68 ps). This is because the Select signal drives not only the gates of the transmission gate transistors but also the input of the internal inverter (used to generate $\bar{S}$).

Table 8: MUX2 Schematic Delay Analysis

Path	Transition	Delay (ps)	Note
Data Path	B: $1 \rightarrow 0$ (S=1, A=1)	9.88	Fast
Data Path	A: $1 \rightarrow 0$ (S=0, B=0)	9.87	Fast
Data Path	B: $0 \rightarrow 1$ (S=1, A=1)	10.78	
Data Path	A: $0 \rightarrow 1$ (S=0, B=0)	10.75	
Select Path	S: $1 \rightarrow 0$ (A=1, B=0)	11.02	
Select Path	S: $0 \rightarrow 1$ (A=1, B=0)	18.68	**Worst-Case**



(a) Data Path Transition of signal B when both S=1 and A=1.



(b) Worst-Case Transition (Select switching $0 \rightarrow 1$).

Figure 17: MUX2 schematic simulation waveforms.

The post-layout simulation was performed for four different Select signal transition scenarios to identify the true critical path. The results, summarized in Table 9, indicate that the configuration where $A = 1$ and $B = 0$ yields the worst-case delays.

Table 9: MUX2 Post-Layout Delay Analysis (Select Path)

Configuration	Select Transition	Output Transition	Delay (ps)	Note
A=1, B=0	$0 \to 1$	Fall ($1 \to 0$)	24.97	Critical
A=1, B=0	$1 \to 0$	Rise ($0 \to 1$)	25.48	**Worst Case**
A=0, B=1	$0 \to 1$	Rise ($0 \to 1$)	20.26	
A=0, B=1	$1 \to 0$	Fall ($1 \to 0$)	22.61	

The delays for the $A = 1, B = 0$ case (≈ 25 ps) are consistently higher than the $A = 0, B = 1$ case ($\approx 20 - 22$ ps). This discrepancy is attributed to the timing skew between the direct Select signal (S) and the inverted Select signal ($\bar{S}$) generated by the internal inverter.

- When passing a logic '1' (e.g., Input A=1), the PMOS transistor of the TG is the strong driver.
- When passing a logic '0' (e.g., Input B=0), the NMOS transistor is the strong driver.

In the worst-case scenario ($A = 1, B = 0$), the transmission gates rely on control signals that are delayed by the inverter. The additional parasitic capacitance from the layout routing of the S and $\bar{S}$ lines further exacerbates this effect, causing the control signals to slew slower and delaying the transmission gates. Figure 18 displays the transient response for the critical $A = 1, B = 0$ configuration. Table 10 compares the worst-case schematic and extracted delays.

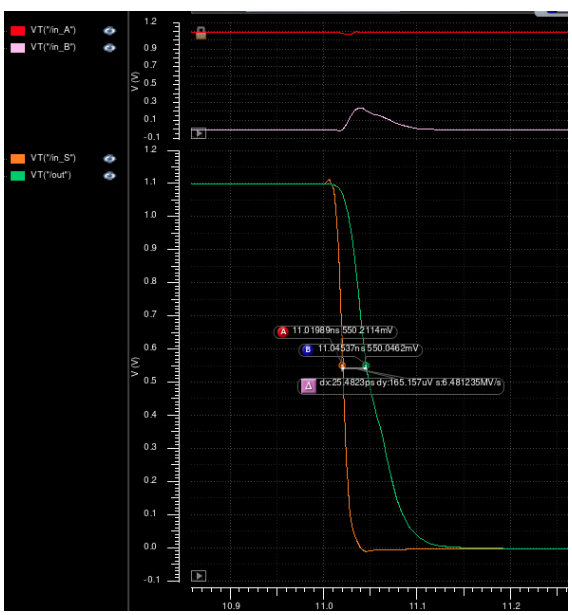


Figure 18: Post-Layout Select $1 \to 0$ (25.48 ps)

Table 10: MUX2 Critical Path Comparison

Metric	Schematic (Worst)	Extracted (Worst)	% Diff
Propagation Delay	18.68 ps	25.48 ps	36.4%